Sum of squared residuals (Error Function):

a quadratic function, leading coefficient > 0 → its derivative = 0 ⟺ its minimum

derivative calculations: $F(a,s) = \sum((\overset{\text{slope}}{a}x_i + s) - y_i)^2 \implies \frac{\partial F}{\partial s} = \sum 2((ax_i + s) - y_i) \quad \& \quad \frac{\partial F}{\partial a} = \sum 2x_i((ax_i + s) - y_i)$

step calculations: $s_{n+1} = s_n - \frac{\partial F}{\partial s}\Big|_{s=s_n} \cdot \eta \quad \& \quad a_{n+1} = a_n - \frac{\partial F}{\partial a}\Big|_{a=a_n} \cdot \eta$



prediction model: linear regression line $\implies$ Predicted Traffic Volume $= a \cdot$ Hour of Day $+ s$

Data Source: metro_Interstate_traffic_volume (Westbound traffic in Minneapolis-St Paul, MN, from 2012-2018)

link: https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume

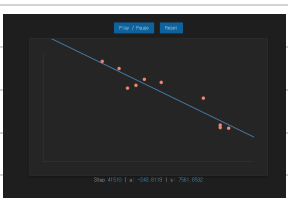
⟹ the link had 48205 data; I chose 10 data points chosen under similar weather conditions to isolate the relationship between solely time and traffic volume

Extracted Pairs: (Time of Day (x_i) vs. Traffic Volume (y_i)):

Traffic Volume Data

Time of Day	Traffic Volume
07:00	5,972
09:00	5,545
10:00	4,378
11:00	4,545
12:00	4,902
14:00	4,721
19:00	3,783
21:00	2,304*
22:00	1,992

Results:



slope: -243.8119 **vertical-intercept: 7561.8532**

Reflection:

Calculus allows machine learning methods to perform an optimized search rather than a brute-force search. By observing the degree of change in error as the program adjusts the parameter, it can move in the right direction more easily. This clever shortcut helps save an immense amount of computing power as it allows the AI to hit the best solution quickly and efficiently with less power.